

Kellen D. Lawson

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Research Interests

- ▷ Exoplanet formation & evolution
- ▷ Coronagraphy
- ▷ Software development
- ▷ Circumstellar disks
- ▷ Polarimetry
- ▷ GPU computing
- ▷ High-contrast imaging
- ▷ Integral field spectroscopy
- ▷ Numerical optimization

Education

Doctor of Philosophy (Physics) — Univ. of Oklahoma, 2016 – 2022

(Advisor: John Wisniewski)

Bachelor of Science (Astrophysics) — College of Charleston, 2010 – 2014

(Advisor: Joseph Carson)

Employment

Assistant Research Scientist (2025 – present)

UMBC / NASA Goddard Space Flight Center

NASA Postdoctoral Program Fellow (2022 – 2025)

NASA Goddard Space Flight Center

Graduate Research Assistant (2018 – 2022)

University of Oklahoma

Graduate Research Assistant (2017 – 2018)

Univ. of Oklahoma & Univ. of Washington

Undergraduate Research Assistant (2013 – 2015)

College of Charleston

Astronomy Software Development

Winnie (Lead Developer): tool for post-processing and analysis of resolved circumstellar disks in JWST observations

SpaceKLIP (Contributing Developer): data reduction pipeline for JWST high-contrast imaging observations

CHARIS DPP – SpecPol Module (Lead Developer): pipeline for Subaru/CHARIS integral field spectropolarimetry data

PyVAN (Lead Developer): tool for assessing variability in large samples of irregularly / sparsely-sampled stellar lightcurves

Grants & Awards

2025 — JWST GO 8826 Funding Award (PI)

JWST Cycle 4 GO

2024 — JWST GO 5298 Funding Award (Co-I)

JWST Cycle 3 GO

2023 — JWST GO 3840 Funding Award (Co-I)

JWST Cycle 2 GO

2022 — NASA Postdoctoral Program Fellowship

NASA Goddard Space Flight Center

2022 — Nielsen Prize

OU Dept. of Physics & Astronomy

2021 — Bullard Dissertation Completion Fellowship

OU Graduate College

2020 — Grants in Aid of Research (PI)

Sigma Xi

2018 — NASA 18-XRP18-0117 (Co-I)

NASA XRP 2018

2018 — TESS GO11264 (Co-I)

TESS Cycle 1 Guest Investigator Program

2014 — Research Presentation Grant (PI)

College of Charleston

2014 — Major Academic Year Support Grant (PI)

College of Charleston

2014 — Dunlap Institute Summer School Tuition & Travel Grant (PI)

Univ. of Toronto

2014 — Richard Petit Award for Outstanding Undergraduate Research

Sigma Xi

2014 — Summer Undergraduate Research with Faculty Grant (PI)

College of Charleston

2013 — Major Academic Year Support Grant (PI)

College of Charleston

Observing Programs

2025 — JWST GO 8826 (PI)

JWST Cycle 4 GO

“The Turning Tide: Discovering Unprecedented Exoplanets and Disks Around Fast-Rotating M Dwarfs”

2025 — Subaru S25A-026 (Co-I)

Subaru Normal Program 2025A

“Planet-Forming Disks around M-type stars in Taurus”

2025 — Subaru S25A-TE239-K (Co-I)

Subaru-Gemini Time Exchange 2025A

“Mapping Europa’s Surface In Polarized Light”

2024 — JWST GO 5298 (Co-I)

JWST Cycle 3 GO

“Catching a cat by the tail: Tracing Dust Dynamics in the Beta Pictoris Debris Disk in the Aftermath of ...”

2023 — Subaru S23A-038 (Co-I)

Subaru Normal Program 2023A

“Characterization of a New, Benchmark Directly-Imaged Exoplanet with SCEXAO-CHARIS”

- 2023** — Subaru S23A-073 (Co-I) Subaru Normal Program 2023A
“Second-Epoch Confirmation of New Directly-Imaged Planet Candidates Around Accelerating Stars”
- 2023** — JWST GO 3840 (Co-I) JWST Cycle 2 GO
“JWST NIRCcam Confirmation of the First Directly Imaged Sub-Saturn Mass Exoplanet”
- 2022** — JWST ERS 1386 (Co-I) JWST Early Release Science
“High Contrast Imaging of Exoplanets and Exoplanetary Systems with JWST”
- 2022** — Subaru S22B-027 (Co-I) Subaru Normal Program 2022B
“Characterizing the Planetary System Around AB Aurigae”
- 2022** — Subaru S22A-TE004-G (Co-I) Subaru-Gemini Time Exchange 2022A
“A Survey of Volume-Limited Herbig Disks in Polarized Light”
- 2022** — Subaru S22A-063 (Science PI) Subaru Normal Program 2022A
“Diagnosing Dynamical Sculpting in Asymmetrical Disks with CHARIS”
- 2021** — Subaru S21B-TE144-G (PI) Subaru-Gemini Time Exchange 2021B
“The Origins of Spiral-Arm Structures in Young Protoplanetary Disks”

Professional Service

- HWO Concept of Operations and Post-Processing TAG** Focus Group Co-lead (2024 – present)
- JWST Users Group at Goddard** Organizer (2024 – present)
- GSFC Exoplanet Seminar Series** Organizer (2023 – present)
- ExoPAG SAG 23: The Impact of Exo-Zodiacal Dust on Exoplanet Direct Imaging Surveys** Team Member (2023)
- JWST ERS Program for Direct Observations of Exoplanetary Systems** Team Member (2022 – present)
- NASA’s Exoplanets Research Program** Review Panelist (c.2020)
- Astronomy & Astrophysics** Journal Referee (c.2020)

Publications

▷ Refereed Publications

- Lawson**, et al. 2025, “*JWST NIRCcam Coronagraphy of the HR 4796A Debris Disk at 2 and 4.4 μm ,*” in prep.
- Lawson**, et al. 2025, “*Winnie: PSF-subtraction, Forward-modeling, and Deconvolution of JWST Coronagraphy of Circumstellar Disk Systems,*” in prep.
- Palatnick, Millar-Blanchaer, ... **Lawson**, et al. 2025, “*JWST NIRCcam Discovery of an M-dwarf Debris Disk in TWA,*” submitted to AJ
- Lazzoni, Bendahan-West, ... **Lawson**, et al. 2025, “*HD 92945 through the eyes of JWST,*” submitted to A&A
- Millar-Blanchaer, Choquet, **Lawson**, et al. 2025, “*The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems VII: Evidence for radially evolving icy grains in the HD 141569A disk via NIRCcam coronagraphic imaging,*” AJ, in press
- Crotts, Carter, **Lawson**, et al. 2025, “*Follow-Up Exploration of the TWA 7 Planet-Disk System with JWST NIRCcam,*” ApJL, 987, L41
- Bogat, Schlieder, **Lawson**, et al. 2025, “*Probing the Outskirts of M Dwarf Planetary Systems with a Cycle 1 JWST NIRCcam Coronagraphy Survey,*” AJ, 170, 225
- Ray, ... **Lawson**, et al. 2025, “*The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems III: Aperture Masking Interferometric Observations of the star HIP 65426 at 3.8 μm ,*” ApJL, 983, L25
- Balmer, ... **Lawson**, et al. 2025, “*JWST-TST High Contrast: Living on the Wedge, or, NIRCcam Bar Coronagraphy Reveals CO₂ in the HR 8799 and 51 Eri Exoplanets’ Atmospheres,*” AJ, 169, 209
- Dykes, Currie, **Lawson**, et al. 2024, “*SCEXAO/CHARIS Near-infrared Scattered-light Imaging and Integral Field Spectropolarimetry of the AB Aurigae Protoplanetary System,*” ApJ, 977, 172
- Chen, **Lawson**, Brandt, et al. 2024, “*Multiband polarimetric imaging of HD 34700 with SCEXAO/CHARIS,*” MNRAS, 533, 2473

- Kammerer, **Lawson**, Perrin, et al. 2024, “*JWST-TST High Contrast: JWST/NIRCam Observations of the Young Giant Planet β Pic b*,” *AJ*, 168, 51
- Vincent, **Lawson**, Currie, et al. 2024, “*SCEXAO/CHARIS Multiwavelength High-contrast Imaging of the BD+45°598 Debris Disk*,” *AJ*, 168, 6
- Lawson**, Schlieder, Leisenring, et al. 2024, “*JWST / NIRCam Detection of the Fomalhaut C Debris Disk in Scattered Light*,” *ApJL*, 967, L8
- Rebollido, ... **Lawson**, et al. 2024, “*JWST-TST High Contrast: Asymmetries, Dust Populations, and Hints of a Collision in the β Pictoris Disk with NIRCam and MIRI*,” *ApJ*, 167, 69
- Petrus, ... **Lawson**, et al. 2024, “*The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. V. Do Self-consistent Atmospheric Models Represent JWST Spectra? A Showcase with VHS 1256–1257 b*,” *ApJL*, 966, L11
- Sallum, ... **Lawson**, et al. 2024, “*The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems IV: NIRISS Aperture Masking Interferometry Performance and Lessons Learned*,” *ApJL*, 963, L2
- Wagner, ... **Lawson**, et al. 2023, “*Direct images and spectroscopy of a giant protoplanet driving spiral arms in MWC 758*,” *Nature Astronomy*
- Lawson**, Schlieder, Leisenring, et al. 2023, “*JWST / NIRCam Coronagraphy of the Young Planet-hosting Debris Disk AU Microscopii*,” *AJ*, 166, 150
- Miles, ... **Lawson**, et al. 2023, “*The JWST Early-release Science Program for Direct Observations of Exoplanetary Systems II: A 1 to 20 μ m Spectrum of the Planetary-mass Companion VHS 1256-1257 b*,” *ApJL*, 947, L6
- Carter, ... **Lawson**, et al. 2023, “*The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems I: High-contrast Imaging of the Exoplanet HIP 65426 b from 2 to 16 μ m*,” *ApJL*, 951, L20
- Currie, ... **Lawson**, et al. 2023, “*Direct imaging and astrometric detection of a gas giant planet orbiting an accelerating star*,” *Science*, 380, 198–203
- Lawson**, Currie, Wisniewski, et al. 2022, “*Constrained Reference Star Differential Imaging: Enabling High-fidelity Imagery of Highly Structured Circumstellar Disks*,” *ApJL*, 935, L25
- Kuzuhara, ... **Lawson**, et al. 2022, “*Direct-imaging Discovery and Dynamical Mass of a Substellar Companion Orbiting an Accelerating Hyades Sun-like Star with SCEXAO / CHARIS*,” *ApJL*, 934, L18
- Currie, **Lawson**, Schneider, et al. 2022, “*Images of embedded Jovian planet formation at a wide separation around AB Aurigae*,” *Nature Astronomy*
- Uyama, Ruane, **Lawson**, et al. 2022, “*A Spatially-resolved Large Cavity of the J0337 Protoplanetary Disk in Perseus*,” *AJ*, 163, 204
- Lawson**, Currie, Wisniewski, et al. 2021, “*Multiband imaging of the HD 36546 debris disk: a refined view from SCEXAO / CHARIS*,” *AJ*, 162, 293
- Currie, ... **Lawson**, et al. 2020, “*SCEXAO / CHARIS Direct Imaging Discovery of a 20 au Separation, Low-mass Ratio Brown Dwarf Companion to an Accelerating Sun-like Star*,” *ApJL*, 904, L25
- Lawson**, Currie, Wisniewski, et al. 2020, “*SCEXAO / CHARIS Near-IR Integral Field Spectroscopy of the HD 15115 Debris Disk*,” *AJ*, 160, 163
- Schutte, **Lawson**, Wisniewski, et al. 2020, “*Discovery of a Nearby Young Brown Dwarf Disk*,” *AJ*, 160, 156
- Silverberg, ... **Lawson**, et al. 2020, “*Peter Pan Disks: Long-lived Accretion Disks Around Young M Stars*,” *ApJ*, 890, 106
- Lawson**, Wisniewski, Bellm, Kowalski, & Shupe 2019, “*Identification of Stellar Flares Using Differential Evolution Template Optimization*,” *AJ*, 158, 119
- Blunt, ... **Lawson**, et al. 2019, “*Radial Velocity Discovery of an Eccentric Jovian World Orbiting at 18 au*,” *AJ*, 158, 181
- Wisniewski, ... **Lawson**, et al. 2019, “*High-fidelity Imaging of the Inner AU Mic Debris Disk: Evidence of Differential Wind Sculpting?*,” *ApJL*, 883, L8
- Currie, ... **Lawson**, et al. 2019, “*No Clear, Direct Evidence for Multiple Protoplanets Orbiting LkCa 15: LkCa 15 bcd are Likely Inner Disk Signals*,” *ApJL*, 877, L3

▷ **Conference Proceedings & White Papers**

- Lawson, Ren, et al.** 2025, “*Telemetry-Aided Post-Processing with Machine Learning Methods for the Habitable Worlds Observatory Coronagraph*,” HWO Coronagraph Concept of Operations and Post-processing Technical Assessment Group
- Wisniewski, ... **Lawson, et al.** 2022, “*UV spectropolarimetry with Polstar: protoplanetary disks*,” *Astrophysics and Space Science*, 367, 12, 122
- Lawson, Currie, Wisniewski, et al.** 2021, “*High-contrast integral field spectropolarimetry of planet-forming disks with SCEXAO / CHARIS*,” *Proc. SPIE* 11823, 118230D
- Currie, ... **Lawson, et al.** 2021, “*A new type of exoplanet direct imaging search: a SCEXAO / CHARIS survey of accelerating stars*,” *Proc. SPIE* 11823, 1182304
- Currie, ... **Lawson, et al.** 2020, “*On-sky performance and recent results from the Subaru coronagraphic extreme adaptive optics system*,” *Proc. SPIE* 11448, 114487H

Selected Presentations

▷ Invited

- “*Telemetry-Aided Post-Processing with Machine Learning Methods for the Habitable Worlds Observatory Coronagraph*,” HWO Coronagraph Concept of Operations and Post-processing Technical Interchange Meeting, Apr 2025
- “*JWST Coronagraphy of Debris Disk Systems*,” Carnegie EPL Astronomy Seminar, Apr 2025
- “*Coronagraphic JWST/NIRCam Images of a Young Planet-Hosting Debris Disk System*,” AAS 241 Press Conference, Jan 2023
- “*Coronagraphic JWST/NIRCam Images of a Young Planet-Hosting Debris Disk System*,” NASA Sciences and Exploration Directorate Director’s Seminar, Dec 2022
- “*The Lord of the Rings: Understanding Planet Formation Through Imaging of Planet-Forming Disks*,” GSFC Exoplanet Seminar Series, Nov 2022
- “*Direct Imaging of Exoplanetary Systems with JWST: On-sky Capabilities & Early Results*,” SPICE Community Workshop, Sep 2022
- “*High-contrast integral field spectropolarimetry of planet-forming disks with SCEXAO/CHARIS*,” University of Kansas Astronomy and Space Physics Seminar, Nov 2021

▷ Contributed

- “*JWST Coronagraphy of Debris Disk Systems: Early Results and Lessons Learned*,” Chesapeake Bay Area Exoplanet Meeting, May 2024
- “*First Scattered Light Detection of the Fomalhaut C Debris Disk with JWST/NIRCam*,” Dust Devils - Debris Disks in the Sonoran Desert, March 2024
- “*High-contrast Polarimetry as a Complement for Total Intensity Circumstellar Disk Imaging*,” ExoPAG 25, Jan 2022
- “*High-contrast integral field spectropolarimetry of planet-forming disks with SCEXAO/CHARIS*,” SPIE Optical Engineering + Applications, Aug 2021
- “*SCEXAO/CHARIS Near-IR Integral Field Spectroscopy of the HD 15115 Debris Disk*,” AAS 236, Jun 2020

Selected Outreach

Speaker: NASA’s Universe of Learning Science Briefing (2023) — Presented a summary of our JWST study of the AU Mic debris disk to science educators.

Member: Lunar Sooners (Univ. of Oklahoma; 2016 – 2022) — A student organization introducing under-served Oklahoma communities to astronomy using a portable planetarium, public telescope observing, discussion panels, and demonstrations.

Event Host: SW OKC Public Library astronomy day (2019) — Astronomy demos and Q&A for children

Event Host: Planetarium at Jay Elementary – Jay, Oklahoma (2019) — Educational event using a portable planetarium

Event Host: Boys and Girls Club of Norman (2018) — Astronomy demonstrations for K-12 students